

Assignment Summary

Delta Lake MERGE Implementation

Objective

The main objective of this assignment was to understand how Delta Lake performs incremental data processing using the MERGE operation. I also learned how to clean data, update existing records, insert new records, and validate the final output.

Dataset Used

For this assignment, I used the **Sample Superstore Dataset**. The dataset was uploaded into Databricks and converted into a Delta table for further processing.

Tools Used

- Databricks Free Edition
- Apache Spark (PySpark)
- Delta Lake
- Python

Work Performed

1. Data Loading

First, I uploaded the dataset into Databricks and loaded it into a Delta table. I displayed the dataset and checked its schema before performing any operations.

2. Data Cleaning

I checked the dataset for null values and duplicate records. No null values were found, and there were no duplicate records in the dataset.

3. Incremental Dataset Creation

To simulate incremental data, I created a second dataset. I updated the **Sales** value for a few existing records and added one new record. This dataset was later used in the MERGE operation.

4. Delta Lake MERGE

I used the Delta Lake MERGE operation to combine the incremental dataset with the main Delta table. Existing records were updated based on **Order ID**, and the new record was inserted successfully.

5. Validation

After the MERGE operation, I validated the final dataset by checking the total number of records and reviewing duplicate records. This helped me verify that the MERGE operation was completed successfully.

6. Final Output

Finally, I displayed the updated Delta table and confirmed that both the updated records and the new record were present in the final dataset.

Result

The assignment was completed successfully. The dataset was loaded into a Delta table, cleaned, merged with incremental data, validated, and displayed successfully. The MERGE operation worked as expected by updating existing records and inserting a new record.

Conclusion

Through this assignment, I learned how Delta Lake supports incremental data processing using the MERGE operation. I also gained hands-on experience with Delta tables, data cleaning, validation, and working with PySpark in Databricks. This assignment helped me understand how incremental updates are handled in real-world data engineering projects.